Digital Design 101 | 2026-1

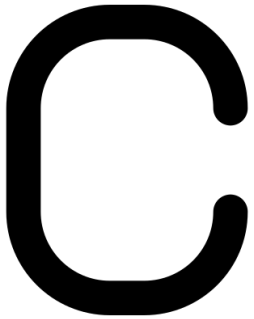
Session6 : Behavioral Modeling

1. Hardware Modeling

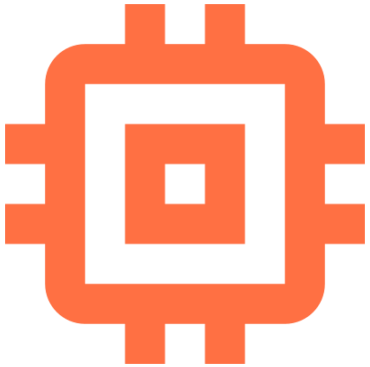
THINK

? SW개발 언어와 HW개발 언어는 어떤 차이점이 있을까?

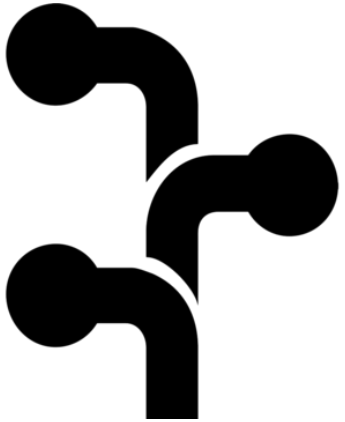
Software Language VS Hardware Descriptive Language

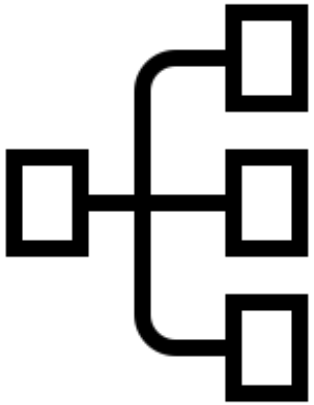


Sequential programming language



Concurrent programming language

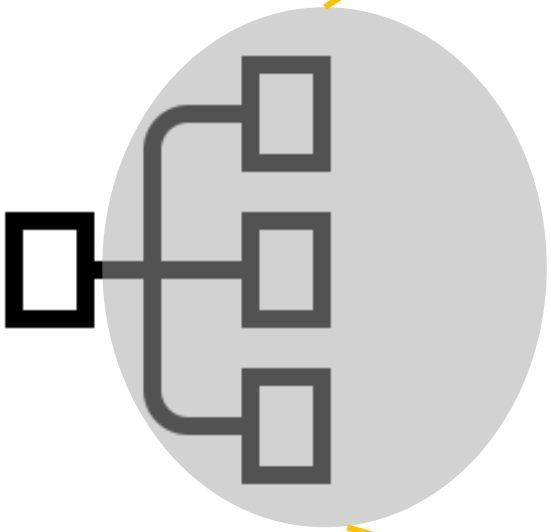




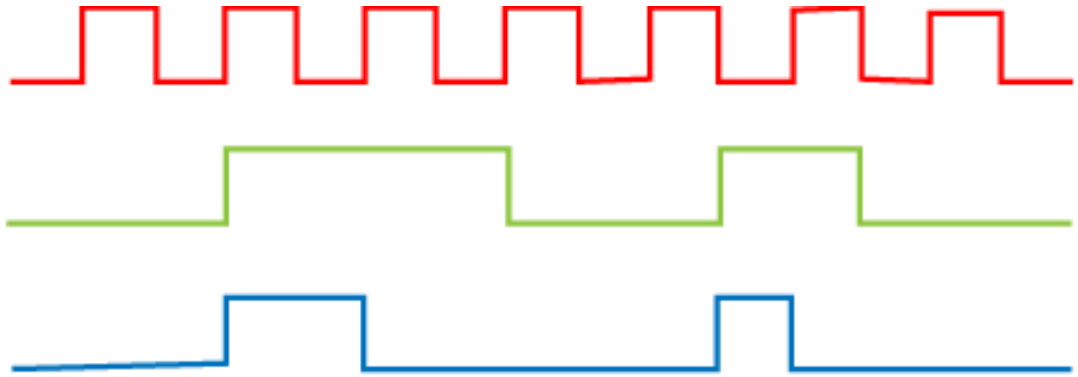
THINK

How can we describe HW's concurrent behavior?

Procedural Block



Procedural Block[s]

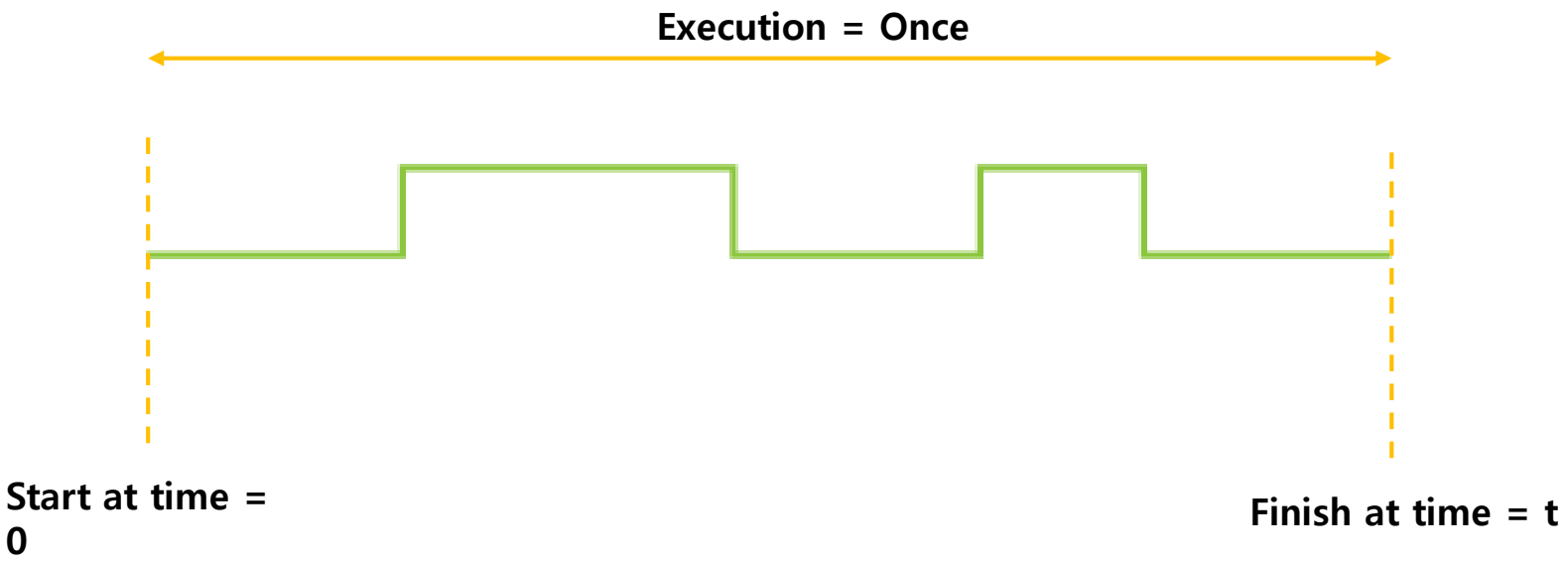


Sperate Activity Flow

시작과 끝이 있는 것(initial) vs 무한히 반복하는 것(always)

2. Behavioral Modeling

Initial Statement: overview

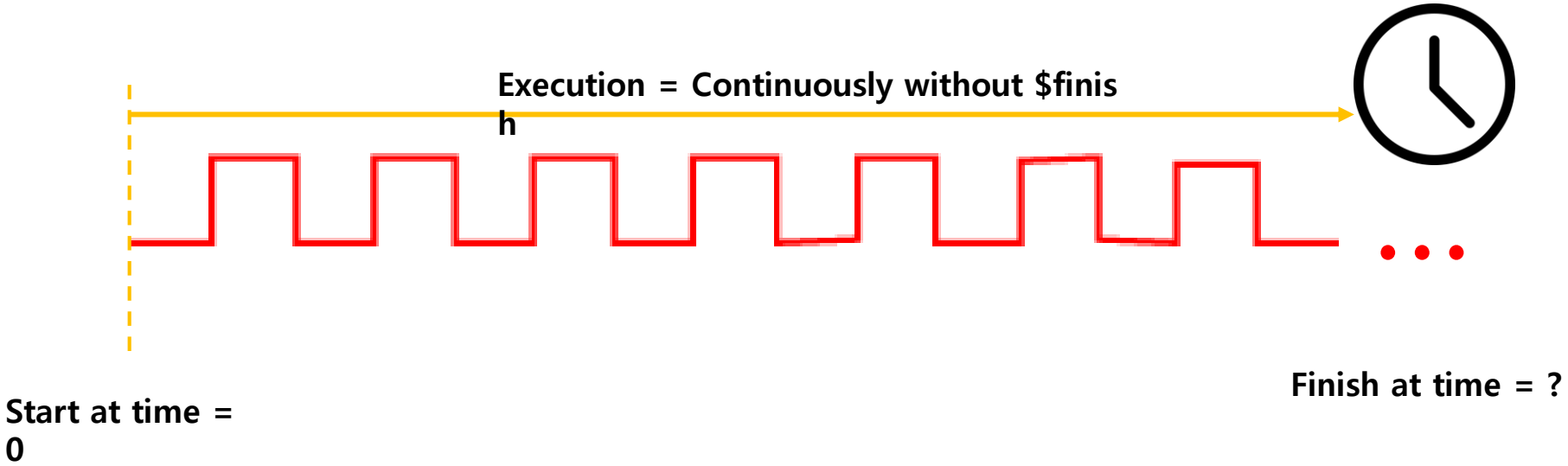


Initial Statement: Example

```
1   module tb_initial;
2
3       reg x, y, a, b, m;
4
5       // 1st block
6       initial
7           m = 1'b0;
8
9       // 2ed block
10      initial
11      begin
12          #5 a = 1'b1;
13          #25 b = 1'b0;
14      end
15
16      // 3rd block
17      initial
18      begin
19          #10 x = 1'b0;
20          #25 y = 1'b1;
21      end
22
23      // 4th block
24      initial
25          #50 $finish;
26
27  endmodule
```

Time at	Statement executed	Block executed
0	m = 1'b0	1
5	a = 1'b1	2
10	x = 1'b0	3
30	b = 1'b0	2
35	y = 1'b1	3
50	$finish	4

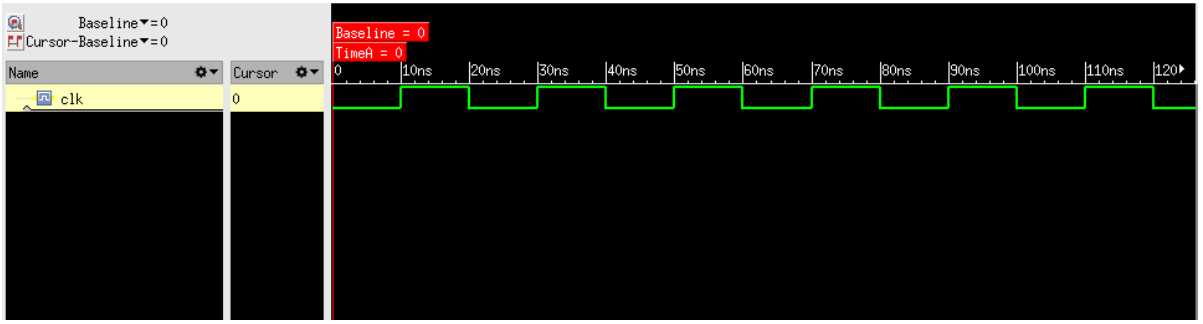
Always Statement: overview



… "always" block is similar to an infinite loop in C language …

Always Statement: Example

```
1   module clock_gen;
2
3   reg clk;
4
5   // 1. Initialize clk at time = 0
6   initial begin
7       clk = 1'b0;
8   end
9
10  // 2. Toggle clk every half-cycle to produce time period = 20
11  always
12      #10 clk = ~clk;
13
14  // 3. Finish system tick
15  initial
16      #1000 $finish;
17
18  endmodule
```



Always Statement: Event control

```
always @ (sensitivity list)
begin
    event(logic statement)
end
```

Always(continuously) event triggered at(@) sensitivity list

→ Sensitivity 조건을 만족할 때만 event를 계속해서 수행

- **Multiple List**

→ always @(a, b, …, n) / always @(a or b or … n)

- **Edge Trigger**

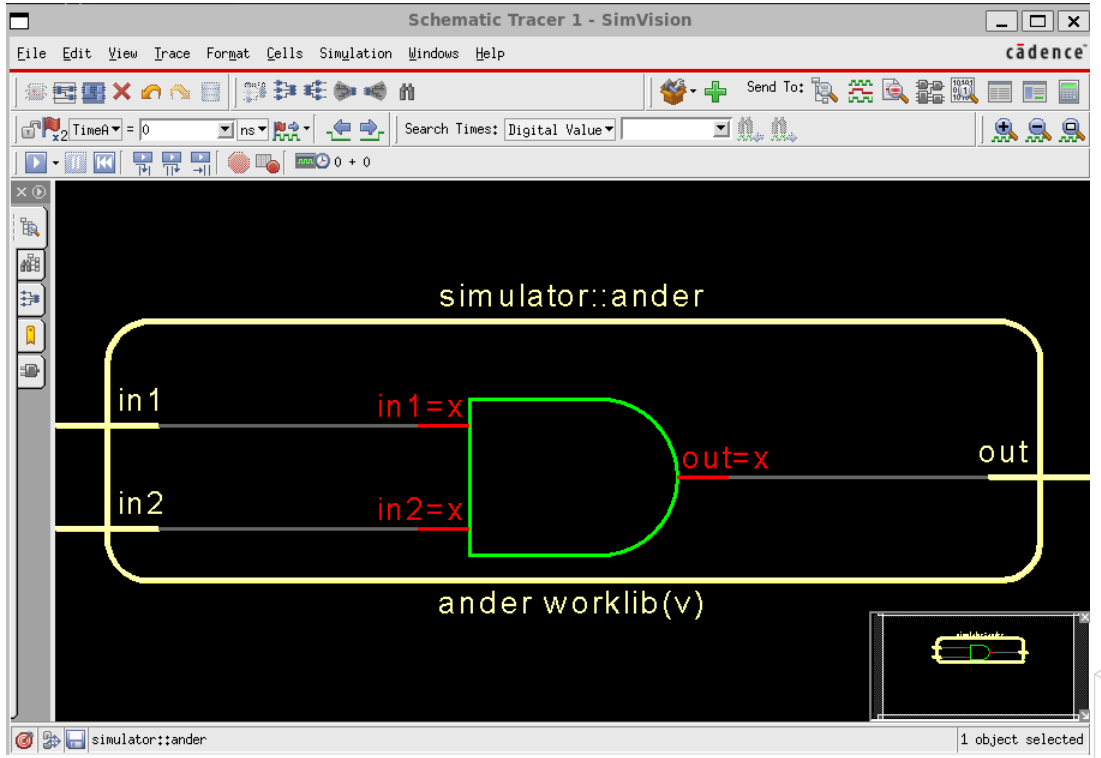
→ always @(posedge/negedge sig)

- **Wild Card**

→ always @* / always @(*)

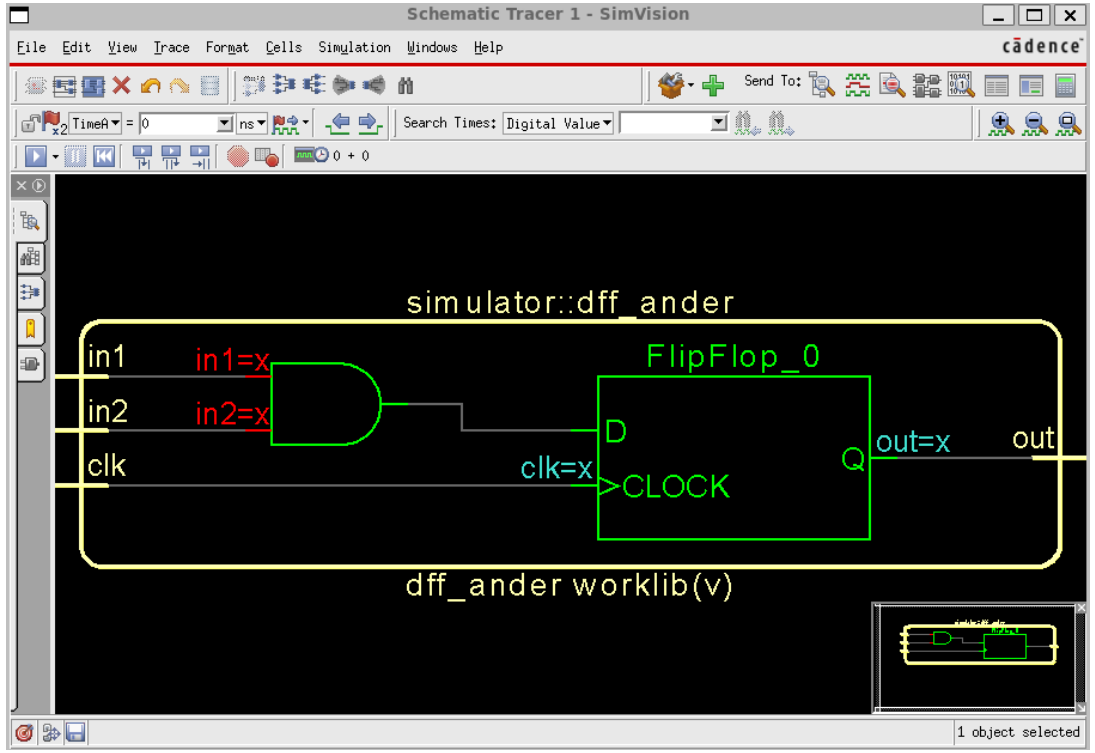
Always Statement: AND gate modeling1

```
1  module ander (
2      in1, in2, out
3  );
4
5  input wire in1, in2;
6  output reg out;
7
8  always @(in1 or in2)
9  begin
10     out = in1 & in2;
11 end
12
13 endmodule
```



Always Statement: AND gate modeling2

```
1   module dff_ander (
2       in1, in2, out, clk
3   );
4
5   input wire in1, in2, clk;
6   output reg out;
7
8
9   always @(posedge clk)
10  begin
11      out = in1 & in2;
12  end
13
14
15
16  endmodule
```



Procedural Assignment

initial | always @ (sensitivity list)

begin

Procedural block

event(logic statement)

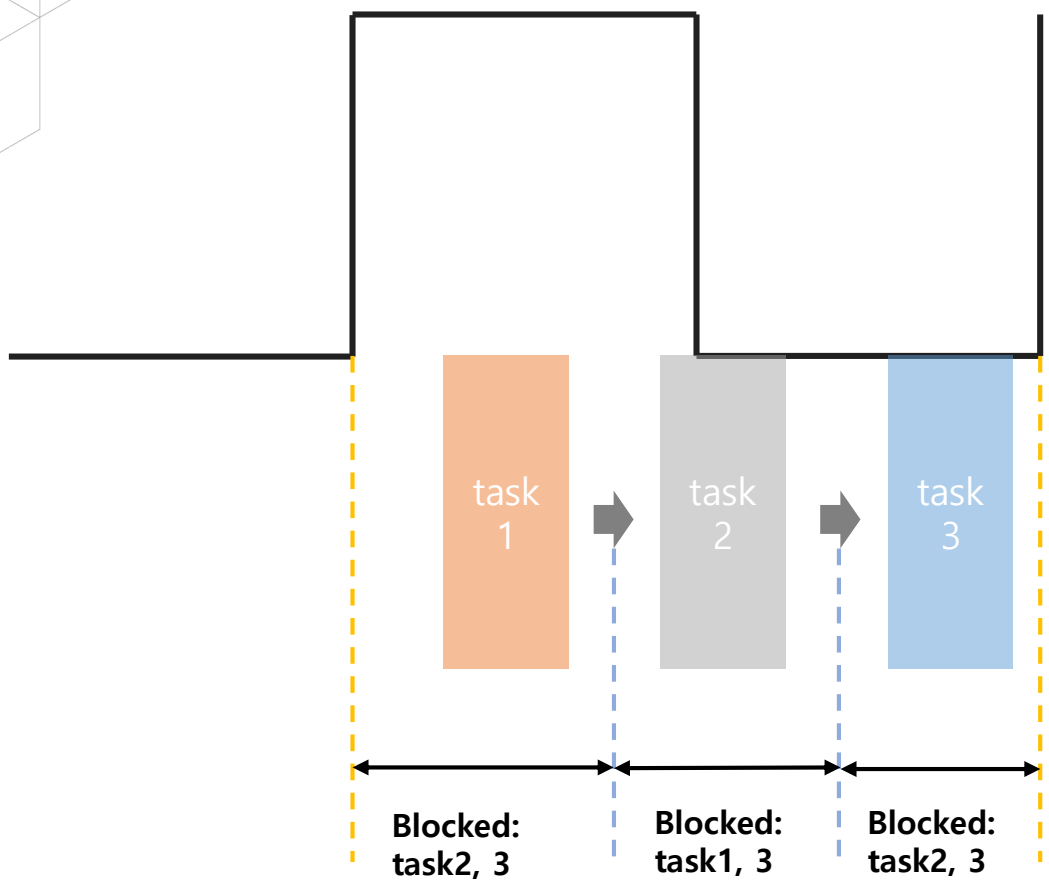
end

Blocking vs Non-blocking

= <=

- **Type of LHS = reg**
- **또 다른 procedural assignment가 변수의 값을 업데이트 하기 전까지는 값을 유지(∴ reg)**

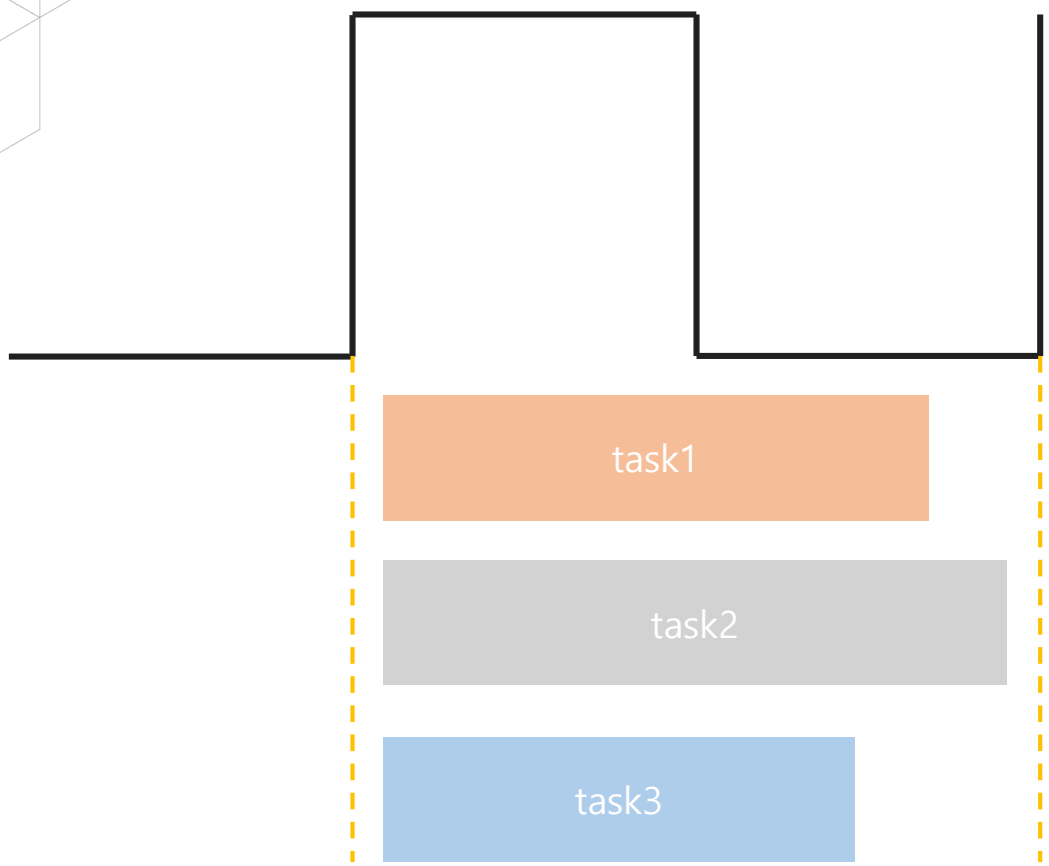
Blocking assignment




```
always @(posedge clk, negedge n_rst)
begin
    b = a; // task #1
    c = b; // task #2
    a = c; // task #3
end
```

… executed in the order tasks are specified in a sequential block …

Non-Blocking assignment



Each task doesn't wait for the others to complete execution!

```
always @(posedge clk, negedge n_rst)
begin
// each task simultaneously occurs at positive clk edge
    b <= a; // task #1
    c <= b; // task #2
    a <= c; // task #3
end
```

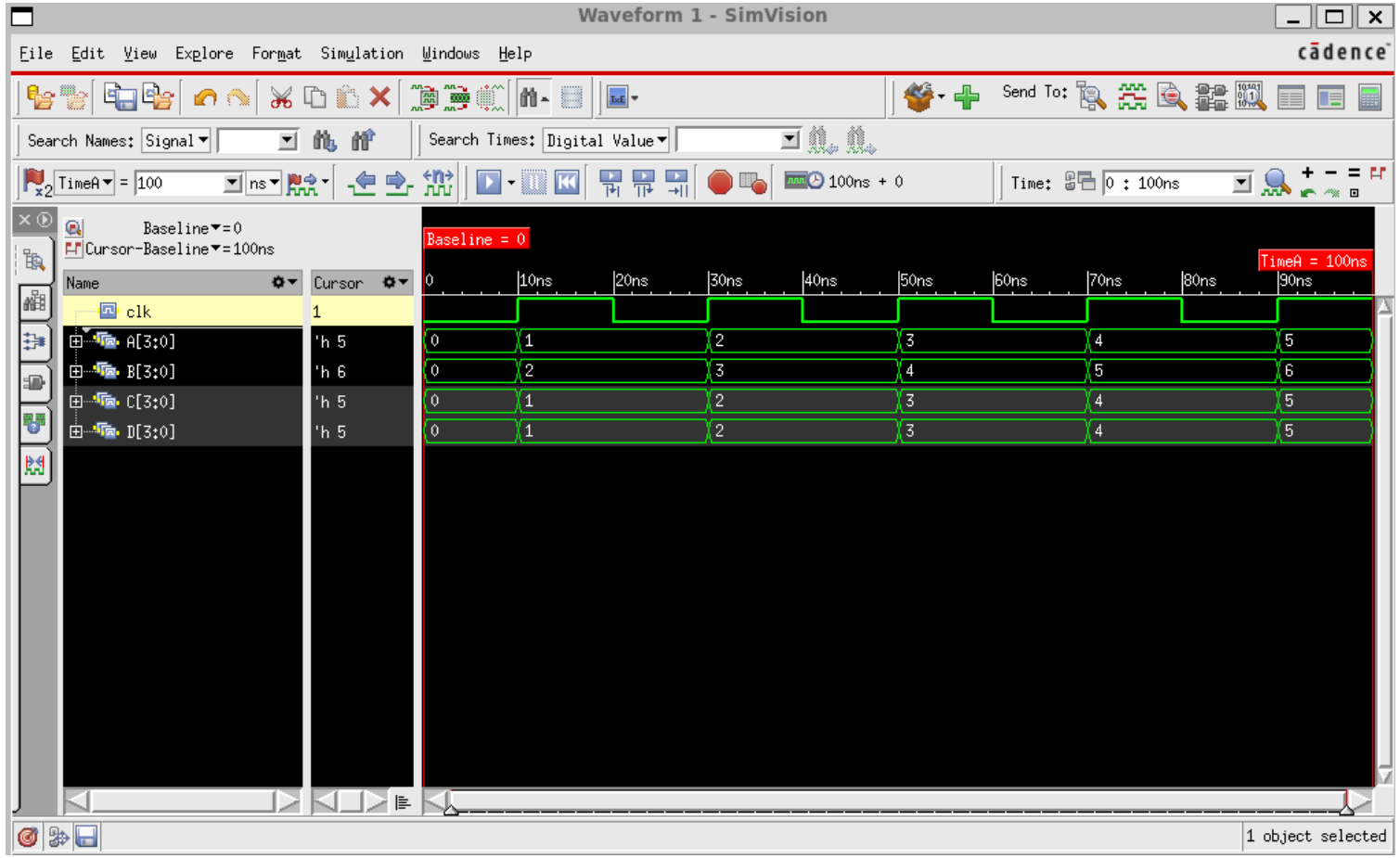
Non-blocking is used to deliver/assign multiple value at the same time

Blocking vs Non-Blocking

```
module non_blocking;

reg [3:0] A;
reg [3:0] B;
reg [3:0] C;
reg [3:0] D;

reg clk;

// time = 0
initial
    clk = 1'b0;

initial begin
    A = 4'd0; B = 4'd0; C = 4'd0; D = 4'd0;
end

// 20ns clock
always
    #10 clk = ~clk;


always @(posedge clk)
begin
    A = A + 1; $display("[%0t] A = 0x%4h", $time, A);
    B = A + 1; $display("[%0t] B = 0x%4h", $time, B);
end


always @(posedge clk)
begin
    C <= C + 1; $display("[%0t] C = 0x%4h", $time, C);
    D <= C + 1; $display("[%0t] D = 0x%4h", $time, D);
end

initial // 5 tick
    #100 $finish;

endmodule
```

```
xcelium> run
[10] A = 0x0001
[10] B = 0x0002
[10] C = 0x0000
[10] D = 0x0000
[30] A = 0x0002
[30] B = 0x0003
[30] C = 0x0001
[30] D = 0x0001
[50] A = 0x0003
[50] B = 0x0004
[50] C = 0x0002
[50] D = 0x0002
[70] A = 0x0004
[70] B = 0x0005
[70] C = 0x0003
[70] D = 0x0003
[90] A = 0x0005
[90] B = 0x0006
[90] C = 0x0004
[90] D = 0x0004
Simulation complete via $finish(1) at time 100 NS + 0
```

Blocking vs Non-Blocking



Conditional Statement: if-(if)else

```
...

if (number_queue < MAX_QUEUE_DEPTH)
begin
    data_queue = data;
    num_dueue = num_dueue + 1;
end
else
begin
    $display("Data queue has been full. Try again");
end

...

if (alu_control == 0)
    out = in1 + in2;
else if (alu_control == 1)
    out = in1 - in2;
else if (alu_control == 2)
    out = in1 * in2;
else // exception handling
    $display("Wrong control signal!!!");

...
```

많은 if-else if 구문이 중첩되어 있다면 어떤 문제가 있을까?

어떤 모듈로 합성 될까?

Conditional Statement: case

```
1   module mux4_to_1 (
2       out,
3       in0, in1, in2, in3,
4       sel1, sel0
5   );
6
7   output reg out;
8   input wire in0, in1, in2, in3;
9   input wire sel1, sel0;
10
11
12  always @(*)
13      if ({sel1, sel0} == 2'd0)
14      out = in0;
15      else if ({sel1, sel0} == 2'd1)
16      out = in1;
17      else if({sel1, sel0} == 2'd2)
18      out = in2;
19      else if({sel1, sel0} == 2'd3)
20      out = in3;
21      else
22      $display("Error!!");
23  endmodule
```

=

```
1   module mux4_to_1 (
2       out,
3       in0, in1, in2, in3,
4       sel1, sel0
5   );
6
7   output reg out;
8   input wire in0, in1, in2, in3;
9   input wire sel1, sel0;
10
11  always @(*) // same as always @(sel0, sel1, in0, in1, in2, in3, in4)
12      case ({sel1, sel0}) // bit concatenantion
13          2'd0 : out = in0;
14          2'd1 : out = in1;
15          2'd2 : out = in2;
16          2'd3 : out = in3;
17          default: $display("Error!!!");
18      endcase
19  endmodule
```

LAB

LAB(1)

- 4X1 MUX Gate Level / Data Flow Design
- 4X1 MUX Behavioral Design

LAB(2)

- Make the counter design that counts 100
  - Counts at positive clock
  - Counts when active-low reset is released
- Verifying DUT using testbench

LAB(3)

- Make the counter design that counts 100
  - Counts at positive clock
  - Counts when active-low reset is released
  - At every 100 cycle, counting 100 should repeats
- Verifying DUT using testbench

LAB(4-1)

- Make 1-bit register referring to the description below:
  - Use positive edge clock
  - Use active low reset
- Verifying DUT using testbench

LAB(4-2)

- Make 4-bit register referring to the description below:
  - Use positive edge clock
  - Use active low reset
- Verifying DUT using testbench

LAB(5)

- Make 10-bit adder referring to the description below:
  - Use positive edge clock
  - Use active low reset
- Verifying DUT using testbench